Battery Modelling:
Mathematics with Physical Meaning

These notes are written for someone who understands mathematics and physics at a graduate level but is not a battery electrochemist. Every equation is preceded by a plain-English explanation of *what physical situation we are trying to describe* and *why we need this particular equation* — and followed by what the equation actually tells us about real battery behaviour.

The story in one paragraph.

A battery stores energy in a chemical form and releases it as electrical current. When current flows, the voltage we measure at the terminals is not the true internal voltage — it is reduced by friction-like losses inside the cell. These losses depend on how much charge is left, how fast current flows, and how hot the cell is. The cell also degrades every time it is used. Our job as modellers is to write down equations that capture all of this so that we can predict voltage, estimate remaining charge, and avoid damaging the cell. That is the thread connecting every equation in this document.

Each section begins with **Why this matters**, states the equation, explains every symbol, and ends with **Physical meaning** and **How this connects forward**.

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The Fundamental Question: Where Does Voltage Come From?

Before any equation, let us establish what a battery actually is from a physics standpoint. A battery consists of two electrodes (anode and cathode) immersed in an electrolyte. At each electrode, a chemical reaction either releases or consumes electrons. The separation of charge between the two electrodes creates a voltage difference — the electric potential that drives current through an external circuit.

This voltage is not fixed. It changes with temperature, with the composition of the electrodes (how much lithium is in them), and with how fast current is flowing. Every equation in this document is, in one way or another, trying to quantify one of these dependencies.

Gibbs Free Energy: The Link Between Chemistry and Voltage

Why this matters. We need to know: given a specific chemical reaction inside the battery, what is the maximum voltage it can produce? This is a thermodynamics question. The answer comes from the Gibbs free energy, which measures the useful work available from a reaction at constant temperature and pressure.

When a chemical reaction occurs inside the battery, it releases free energy ΔG (in joules per mole). If the reaction also moves n electrons through a potential difference E (the cell voltage), then the electrical work done is $n\mathcal{F}E$ joules per mole, where $\mathcal{F}$ is Faraday's constant (the charge of one mole of electrons). Setting these equal:

Gibbs–voltage relation

$$\Delta G = -n\mathcal{F}E$$

ΔG	change in Gibbs free energy of the reaction [$\mathrm{J\,mol^{-1}}$]
n	number of electrons transferred per formula unit
$\mathcal{F} = 96\,485$	Faraday constant [$\mathrm{C\,mol^{-1}}$]
E	cell voltage [V]

Physical meaning. The negative sign means: if the reaction releases energy ($\Delta G < 0$, spontaneous), the cell produces a positive voltage. A larger free-energy release means a higher voltage. This is why lithium-ion cells produce $\sim 3.6\,\mathrm{V}$ while lead-acid cells produce $\sim 2\,\mathrm{V}$ — the lithium chemistry releases more Gibbs free energy per electron transferred.

How this connects forward. The voltage E in this equation is the theoretical maximum under equilibrium. In practice, the electrode composition changes as the cell charges and discharges, so ΔG changes. The Nernst equation (next section) captures how E varies with electrode composition, which is what we call the open-circuit voltage.

Open-Circuit Voltage: What the Battery "Wants" to Be

The Nernst Equation

Why this matters. Imagine you fully charge a lithium-ion cell and then disconnect everything and wait 30 minutes. The voltage you measure is called the *open-circuit voltage* (OCV). It is the equilibrium voltage — what the cell settles to with no current flowing. This voltage changes depending on how much lithium is in the electrodes, which is exactly what the state of charge tracks. The Nernst equation gives us a first-principles formula for how this equilibrium voltage depends on the composition of the electrodes.

For a general reaction with reaction quotient Q_r (a ratio that tracks the current concentrations of reactants and products):

Nernst equation

$$E = E^\circ - \frac{\mathcal{R}T}{n\mathcal{F}} \ln Q_r$$

E	equilibrium voltage at current electrode composition [V]
E°	standard potential (at a defined reference composition) [V]
$\mathcal{R} = 8.314$	universal gas constant [$\mathrm{J\,mol^{-1}\,K^{-1}}$]
T	absolute temperature [K]
n	electrons transferred per reaction event
$\mathcal{F}$	Faraday constant [$\mathrm{C\,mol^{-1}}$]
Q_r	reaction quotient (tracks how far composition has shifted)

Physical meaning. The term $\mathcal{R}T/\mathcal{F} \approx 25.7\,\mathrm{mV}$ at room temperature. This is a small number. It means that even a tenfold change in electrode composition only shifts the voltage by about 59 mV per electron. This is why the OCV–SoC curve of a lithium-ion cell has such long flat plateaux: large changes in lithium content cause only small changes in voltage. The temperature dependence (T in the numerator) tells us that the OCV itself changes slightly with temperature — hotter cells have a slightly different OCV curve.

How this connects forward. In practice, we do not evaluate Q_r analytically for intercalation electrodes — the chemistry is too complex. Instead, we measure the OCV–SoC relationship experimentally and store it as a look-up table. The Nernst equation just explains *why* that curve has the shape it has. The next step is defining the state of charge so we can use that look-up table.

OCV as a Function of State of Charge

Why this matters. We need a single number that tells us "how full" the battery is, and we need a function that maps that number to the equilibrium voltage. State of charge (SoC, symbol z) is that number. Knowing z and the OCV–SoC curve gives us the equilibrium voltage without any current flowing.

OCV look-up

$$V_{\mathrm{oc}} = f(z), \qquad z \in [0, 1]$$

V_{oc}	open-circuit voltage [V]
z	state of charge (0 = empty, 1 = full)
f	measured function, stored as interpolated table

A common analytical fit that captures the shape of an NMC cell:

$$f(z) = a_0 + a_1 z + a_2 z^2 + a_3 z^3 + a_4 \exp\big(a_5(z-1)\big)$$

The exponential term captures the steep voltage rise near $z = 0$ (nearly empty). Parameters $a_0 \dots a_5$ are fitted to measurements for each specific cell.

Physical meaning. Think of $V_{\text{oc}}(z)$ as the battery's "resting" voltage. It is what the voltmeter reads when the battery has been sitting disconnected long enough for everything to settle. A fully charged NMC cell rests near 4.2 V; an empty one rests near 3.0 V. The flat middle region (roughly 3.6–3.9 V) is the plateau where lithium staging transitions occur in the graphite anode.

How this connects forward. V_{oc} is what the battery voltage *would be* if no current were flowing. The moment current flows, additional voltage drops appear inside the cell. These drops are modelled by the equivalent circuit in Section 4. The difference between V_{oc} and the actual terminal voltage tells us how much energy is being wasted as heat.

State of Charge: Tracking What Remains

Why this matters. We need to track how much energy is left in the battery at every moment. State of charge is the battery equivalent of a fuel gauge. Unlike a fuel tank, however, we cannot measure it directly — we can only measure voltage and current. The simplest way to estimate SoC is to integrate the current over time: charge in minus charge out equals remaining charge. This method is called Coulomb counting.

Continuous-Time SoC Equation

SoC dynamics (differential equation)

$$\frac{\mathrm{d}z}{\mathrm{d}t} = -\frac{\eta_c \cdot i(t)}{Q_{\text{nom}}}$$

$z(t)$	state of charge at time t [dimensionless, 0–1]
$i(t)$	current [A]; positive convention = discharge
Q_{nom}	nominal capacity [As] (= Ah rating × 3600)
η_c	coulombic efficiency [≈ 0.999 during charge]

Physical meaning. The equation says: SoC decreases at a rate proportional to current. A cell with $Q_{\text{nom}} = 50$ Ah discharging at 50 A (1 C rate) will deplete from full to empty in exactly 1 hour under this model. The efficiency factor $\eta_c < 1$ during charging accounts for the fact that a small fraction of the charge current goes into side reactions (SEI growth) rather than storing energy — more charge goes in than can come out.

Discrete-Time (Euler) Update

In a real battery management system (BMS), we compute SoC at discrete time steps Δt (typically 0.1 to 10 seconds):

Coulomb counting update (implemented in software)

$$z_{k+1} = z_k - \frac{\eta_c \cdot i_k \cdot \Delta t}{Q_{\text{nom}}}$$

z_k	SoC at time step k
i_k	measured current during step k [A]
Δt	time step duration [s]

Why Coulomb Counting Drifts

Why this matters. If Coulomb counting is so simple, why does every BMS also use voltage-based correction? Because any current sensor has noise: $\tilde{i}_k = i_k + \epsilon_k$, where ϵ_k is random error. Each step adds a small error to z. Over time these errors accumulate. Let us quantify exactly how fast.

If the current noise is i.i.d. with variance σ_i^2, the SoC estimation error variance after k steps is:

Coulomb counting drift (random walk)

$$\mathrm{Var}[\hat{z}_k - z_k] = \left(\frac{\Delta t}{Q_{\mathrm{nom}}}\right)^2 k\,\sigma_i^2 \;\propto\; t$$

$\hat{z}_k$	estimated SoC (from noisy current measurement)
z_k	true SoC
σ_i^2	variance of the current sensor noise [A^2]

Physical meaning. The error grows without bound as time passes ($\propto t$). This is the definition of a *random walk* — it never settles. For a cell with $Q_{\mathrm{nom}} = 50\,\mathrm{Ah}$, a $1\,\mathrm{A}$ sensor noise ($\sigma_i = 1\,\mathrm{A}$) at $\Delta t = 1\,\mathrm{s}$ accumulates an SoC standard deviation of about 1 % every 5 hours. After a day, the estimate could easily be off by 3–5 %. In a real EV, this means the range estimate is wrong.

How this connects forward. This drift is the motivation for Section 8 (Kalman filter). The Kalman filter periodically corrects the drifting Coulomb counter by comparing the model's predicted voltage against the measured voltage. The two methods compensate each other's weaknesses.

Equivalent Circuit Model: Capturing Voltage Losses

Why this matters. We know the resting voltage $V_{\mathrm{oc}}(z)$. But the moment we connect a load and draw current, the terminal voltage drops below V_{oc}. Why? Because current flow inside the battery encounters resistance. There are actually several distinct physical mechanisms, each with a different timescale. The equivalent circuit model (ECM) represents each mechanism as a familiar circuit element, making the mathematics tractable.

The three mechanisms are:

1. **Ohmic resistance** R_0: electrons moving through the current collectors and ions moving through the bulk electrolyte cause an *instantaneous* voltage drop the moment current flows.

2. **Charge-transfer resistance** R_1**, capacitance** C_1: at the electrode–electrolyte interface, ions must cross an energy barrier. This process builds up slowly (timescale

$\tau_1 = R_1 C_1 \sim 10\,\mathrm{s}$) and dissipates slowly when current stops.

3. **Diffusion resistance R_2, capacitance C_2**: lithium diffusing through solid electrode particles is slow. This effect has a longer timescale ($\tau_2 = R_2 C_2 \sim 100\,\mathrm{s}$).

Terminal Voltage Equation

Terminal voltage (what the voltmeter reads)

$$V_\mathrm{t} = V_\mathrm{oc}(z) - iR_0 - V_\mathrm{RC1} - V_\mathrm{RC2}$$

V_t	terminal (measurable) voltage [V]
$V_\mathrm{oc}(z)$	open-circuit voltage from SoC [V]
i	current [A] (positive = discharge)
R_0	ohmic resistance [ohm]
V_RC1	voltage across RC pair 1 (charge-transfer) [V]
V_RC2	voltage across RC pair 2 (diffusion) [V]

Physical meaning. The terminal voltage is the OCV minus three voltage drops. The first (iR_0) is immediate: apply current, voltage instantly drops. The other two ($V_\mathrm{RC1}, V_\mathrm{RC2}$) build up gradually. This is why, if you stop discharging a battery, the voltage immediately jumps (the iR_0 drop disappears) but then slowly recovers over minutes (the RC voltages decay). You have certainly seen this behaviour: a phone battery appears dead, but after resting shows a few percent remaining.

RC Pair State Equation and Its Exact Discretisation

Why this matters. Each RC pair has its own internal voltage state V_ℓ that evolves over time. We need an equation of motion for this state. The physics gives us a first-order ODE. Then, because we implement the model in discrete time (on a microcontroller or in simulation), we need to solve that ODE exactly — not approximately — to avoid numerical instability.

The physics of an RC branch gives:

$$\dot{V}_\ell = -\frac{V_\ell}{\tau_\ell} + \frac{i}{C_\ell}, \qquad \tau_\ell = R_\ell C_\ell$$

Solving this ODE *exactly* over one time step, assuming current is constant within the step (zero-order-hold assumption):

RC branch update (exact discrete-time solution)

$$V_{\ell,k+1} = V_{\ell,k}\, e^{-\Delta t/\tau_\ell} + i_k R_\ell \Big(1 - e^{-\Delta t/\tau_\ell}\Big)$$

$V_{\ell,k}$	RC voltage at step k [V]
$\tau_\ell = R_\ell C_\ell$	time constant of this branch [s]
$e^{-\Delta t/\tau_\ell}$	decay factor per step (between 0 and 1)
$i_k R_\ell$	steady-state value V_ℓ would settle to [V]

Physical meaning. Look at the structure of this equation. It says: the RC voltage at the next step is a mixture of two things — (1) what it was before, faded by the factor $e^{-\Delta t/\tau}$

(the memory of the past), and (2) the current driving it toward a new steady state iR_ℓ (the input forcing it). If the time step Δt is much smaller than τ_ℓ, the decay factor is close to 1 and the voltage changes slowly. If $\Delta t \gg \tau_\ell$, the decay factor is near 0 and the voltage instantly follows the current — we cannot see the dynamics. The Euler approximation (replacing e^{-x} by $1 - x$) becomes numerically unstable for $\Delta t > \tau_\ell$; this exact formula has no such restriction.

How this connects forward. We now have three state equations: one for z, one for V_{RC1}, one for V_{RC2}. Together they form a state-space model that can be plugged directly into a Kalman filter. But first, we need to model how temperature changes (it affects all the resistances) and how the cell ages.

Thermal Model: Why Temperature Matters So Much

Why this matters. Temperature is the single most important external variable in battery behaviour. At low temperature, ions move slowly through the electrolyte — resistance increases, available capacity decreases, and the risk of lithium plating during fast charging rises sharply. At high temperature, the battery ages faster and can enter thermal runaway. We need an equation that predicts how cell temperature evolves during operation.

Lumped Heat Balance

The simplest useful thermal model treats the cell as a single object with uniform temperature. Energy conservation says: the rate of change of stored thermal energy equals heat generated minus heat lost to the environment.

Lumped thermal model

$$C_{\mathrm{th}} \frac{\mathrm{d}T}{\mathrm{d}t} = \dot{Q}_{\mathrm{gen}} - \dot{Q}_{\mathrm{cool}}$$

$$\dot{Q}_{\mathrm{gen}} = i^2 R_{\mathrm{eff}}, \qquad \dot{Q}_{\mathrm{cool}} = \frac{T - T_{\mathrm{amb}}}{R_{\mathrm{th}}}$$

C_{th}	thermal capacitance of the cell [$\mathrm{J\,K^{-1}}$]
T	cell temperature [K]
$\dot{Q}_{\mathrm{gen}}$	heat generated by Ohmic losses [W]
$\dot{Q}_{\mathrm{cool}}$	heat lost to surroundings [W]
$i^2 R_{\mathrm{eff}}$	Joule heating ($R_{\mathrm{eff}} = R_0 + R_1 + R_2$)
R_{th}	thermal resistance to ambient [$\mathrm{K\,W^{-1}}$]
T_{amb}	ambient temperature [K]

Physical meaning. $\dot{Q}_{\mathrm{gen}} = i^2 R_{\mathrm{eff}}$ is just Joule's law: current through resistance produces heat. The *quadratic* dependence on current is critical: doubling the current quadruples the heat. A cell discharging at 2 C generates four times the heat of a 1 C discharge. The cooling term $\dot{Q}_{\mathrm{cool}}$ is Newton's law of cooling: heat flows out in proportion to the temperature difference.

At steady state ($\mathrm{d}T/\mathrm{d}t = 0$), these two terms balance and the cell settles at:

$$T_{\mathrm{ss}} = T_{\mathrm{amb}} + i^2 R_{\mathrm{eff}} R_{\mathrm{th}}$$

For a typical 50 Ah cell with $R_{\mathrm{eff}} = 3\,\mathrm{m\Omega}$ and $R_{\mathrm{th}} = 2\,\mathrm{K\,W^{-1}}$: at 50 A (1 C) the temperature rise is $50^2 \times 0.003 \times 2 = 15\,\mathrm{K}$. At 100 A (2 C) it is 60 K — a potentially dangerous temperature.

How this connects forward. Temperature directly changes all the resistances in the ECM. This coupling between the thermal model and the circuit model is what makes batteries genuinely nonlinear: more current → more heat → lower resistance → more current for the same voltage. The next section quantifies exactly how resistance depends on temperature.

Arrhenius Scaling: How Resistance Changes with Temperature

Why this matters. We established that temperature changes battery resistance, but we have not said *how*. The answer comes from statistical mechanics. Ion transport in a battery (through the electrolyte and across electrode interfaces) involves ions hopping over energy barriers. The fraction of ions with enough thermal energy to make the hop is exponentially sensitive to temperature. This gives rise to the Arrhenius law.

Arrhenius temperature correction

$$R(T) = R(T_{\mathrm{ref}}) \cdot \exp\left[\frac{E_{\mathrm{a}}}{\mathcal{R}}\left(\frac{1}{T} - \frac{1}{T_{\mathrm{ref}}}\right)\right]$$

$R(T)$	resistance at temperature T [Ω]
$T_{\mathrm{ref}} = 298.15\,\mathrm{K}$	reference temperature (25 °C)
E_{a}	activation energy [$\mathrm{J\,mol^{-1}}$]; typically 4000–10000
$\mathcal{R} = 8.314$	gas constant [$\mathrm{J\,mol^{-1}\,K^{-1}}$]

Physical meaning. The ratio $E_{\mathrm{a}}/\mathcal{R}$ is typically 500–1200 K. The argument of the exponential at $T = 253\,\mathrm{K}$ (−20°C) relative to $T_{\mathrm{ref}} = 298\,\mathrm{K}$ is approximately $1000 \times (1/253 - 1/298) \approx 0.60$. So $e^{0.60} \approx 1.82$: the resistance nearly doubles. For $E_{\mathrm{a}} = 10\,000\,\mathrm{J\,mol^{-1}}$ it triples.

This explains why EV range drops dramatically in winter. It is not that the battery loses capacity in the cold in some mysterious way — it is that resistance increases so much that the same current causes larger voltage drops, and the cell hits the minimum voltage cutoff earlier. Less energy can be extracted before the voltage floor is reached.

How this connects forward. $R_0(T)$, $R_1(T)$, and $R_2(T)$ each follow Arrhenius scaling with their own activation energies. This means every parameter in the ECM needs to be evaluated at the current cell temperature before computing the terminal voltage. The thermal model and the ECM are therefore coupled: temperature affects resistance, and current (through resistance) affects temperature.

Capacity Fade: Why Batteries Die

Why this matters. Batteries do not last forever. With every charge–discharge cycle, a small amount of lithium is permanently lost and can no longer participate in the electrochemical reaction. Additionally, over time (even sitting on a shelf), degradation reactions consume electrode material. We need equations that predict how capacity decreases so that we can forecast remaining lifetime.

Why Capacity Fades: The Physics in Brief

During charging, a thin layer called the Solid Electrolyte Interphase (SEI) forms on the graphite anode. The SEI consumes lithium irreversibly. It grows slowly in thickness, and its growth is limited by how fast lithium ions can diffuse through the existing layer. This diffusion-limited growth gives a characteristic $\sqrt{t}$ time dependence.

Semi-Empirical Fade Model

Capacity fade (Wang model)

$$Q_{\text{fade}} = B(T) \cdot A_h^z, \qquad B(T) = B_0 \exp\left(-\frac{E_{\text{a}}^{\text{age}}}{\mathcal{R}T}\right)$$

Q_{fade}	fractional capacity loss [%]
A_h	cumulative Ah throughput (total charge passed) [Ah]
$z \approx 0.55$	power-law exponent (less than 1 due to diffusion limit)
$B(T)$	temperature-dependent prefactor
$E_{\text{a}}^{\text{age}}$	activation energy for ageing reaction

Physical meaning. The exponent $z = 0.55 < 1$ is the fingerprint of the $\sqrt{t}$ SEI growth. If fade were linear, z would be 1; the fact that $z < 1$ means the battery degrades *faster when young* and slows down as the SEI layer thickens and self-passivates. The Arrhenius prefactor $B(T)$ tells us that higher temperatures accelerate ageing exponentially: a battery kept at 40 °C permanently will age several times faster than one kept at 25 °C.

Simplified Linear Cycle Model

For many engineering applications, a simpler first-order approximation is sufficient:

Linear capacity fade

$$Q_{\text{cap}}(N) = Q_{\text{nom}}\big(1 - \alpha_{\text{fade}} N\big), \qquad N_{\text{EOL}} = \frac{0.2}{\alpha_{\text{fade}}}$$

N	number of equivalent full cycles
α_{fade}	capacity loss per cycle [$\approx 10^{-4}$ to 5×10^{-4}]
N_{EOL}	cycle count at end-of-life (80 % of original capacity)

How this connects forward. Capacity fade changes Q_{nom} in the Coulomb counting equation. If the BMS does not update Q_{nom} to reflect the aged capacity, the SoC estimate will be systematically wrong: the battery will appear fuller than it really is, and the range estimate will be optimistic.

The Kalman Filter: Fixing the Drifting SoC

Why this matters. We have two methods to estimate SoC. Coulomb counting (Section 3) is precise over short times but drifts. Voltage-based lookup (invert $V_{\text{oc}}(z)$ to get z from measured voltage) is immune to drift but is very noisy because the OCV curve is flat — a 10 mV measurement error becomes a 2–5 % SoC error. We want to fuse the strengths of both methods: the accuracy of current integration and the stability of voltage correction. The Extended Kalman Filter (EKF) is the principled way to do this.

The idea: at every time step, we have a prediction of V_t from the model. We also have the actual measured V_t. The difference between them (the "innovation" or residual) tells us whether our SoC estimate is too high or too low. The Kalman filter computes how much to correct the estimate, balancing model uncertainty against measurement noise — automatically.

State Vector and System Model

Collect the three internal states into a vector:

$$\mathbf{x}_k = \begin{bmatrix} z_k \\ V_{\text{RC1},k} \\ V_{\text{RC2},k} \end{bmatrix}$$

The model propagates this state forward (using the ECM equations from Section 4) and the measurement is the terminal voltage:

State-space model for the EKF

$$\begin{aligned} \mathbf{x}_{k+1} &= \mathbf{A}\,\mathbf{x}_k + \mathbf{B}\,i_k + \mathbf{w}_k && \text{(predict next state)} \\ y_k &= h(\mathbf{x}_k, i_k) + v_k && \text{(predicted voltage measurement)} \end{aligned}$$

$$h(\mathbf{x}, i) = V_{\text{oc}}(z) - iR_0 - V_{\text{RC1}} - V_{\text{RC2}}$$

$\mathbf{A}, \mathbf{B}$	state transition matrices (from ZOH equations)
$\mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_w)$	process noise: model is imperfect
$v_k \sim \mathcal{N}(0, R_v)$	measurement noise: sensor is imperfect
$y_k = V_{t,k}$	measured terminal voltage

The EKF Recursion

The EKF alternates between two steps at every time instant.

Step 1 — Prediction (run the model forward)

$$\begin{aligned} \hat{\mathbf{x}}_{k|k-1} &= \mathbf{A}\,\hat{\mathbf{x}}_{k-1|k-1} + \mathbf{B}\,i_{k-1} \\ \mathbf{P}_{k|k-1} &= \mathbf{A}\,\mathbf{P}_{k-1|k-1}\,\mathbf{A}^\top + \mathbf{Q}_w \end{aligned}$$

$\hat{\mathbf{x}}_{k|k-1}$	predicted state before seeing measurement at k
$\mathbf{P}_{k|k-1}$	predicted covariance (how uncertain we are)

Step 2 — Update (correct with the measurement)

$$\mathbf{C}_k = \left[\frac{\mathrm{d}V_{\mathrm{oc}}}{\mathrm{d}z}\bigg|_{\hat{z}},\ -1,\ -1 \right] \quad \text{(Jacobian: linearise } h \text{ around current estimate)}$$

$$\mathbf{K}_k = \mathbf{P}_{k|k-1}\,\mathbf{C}_k^\top \left(\mathbf{C}_k\,\mathbf{P}_{k|k-1}\,\mathbf{C}_k^\top + R_v \right)^{-1}$$

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k \underbrace{\left(y_k - h(\hat{\mathbf{x}}_{k|k-1}, i_k) \right)}_{\text{innovation: measured}-\text{predicted}}$$

$$\mathbf{P}_{k|k} = (\mathbf{I} - \mathbf{K}_k\,\mathbf{C}_k)\,\mathbf{P}_{k|k-1}$$

$\mathbf{K}_k$	Kalman gain: how much to trust the measurement [0–1]
innovation	residual between actual and model-predicted voltage
$\mathbf{P}_{k\|k}$	updated (reduced) uncertainty covariance

Physical meaning. The Kalman gain $\mathbf{K}_k$ is the key. When the model is uncertain ($\mathbf{P}$ is large) relative to measurement noise (R_v), $\mathbf{K}_k$ is large: trust the measurement more, correct aggressively. When the sensor is noisy (R_v is large), $\mathbf{K}_k$ is small: trust the model, correct conservatively.

The innovation (measured minus predicted voltage) is the signal that drives the SoC correction. If the cell is producing less voltage than predicted, it is probably less charged than estimated, and the filter adjusts $\hat{z}$ downward accordingly.

One important limitation: in the flat OCV plateau, $\mathrm{d}V_{\mathrm{oc}}/\mathrm{d}z \approx 0$, so $\mathbf{C}_k \approx [0, -1, -1]$. The SoC entry vanishes from the Jacobian — voltage measurements carry no information about SoC in this region. The filter reverts to pure Coulomb counting. This is a fundamental observability issue, not a flaw in the implementation.

How Everything Connects: The Complete Model

The following table shows how each equation feeds into the next. Reading this as a flow will help you see the model as a whole rather than a collection of isolated formulae.

Equation	**Inputs**	**Output feeds into**
Coulomb counting	Current i, time step Δt	SoC z
OCV look-up	SoC z	ECM terminal voltage, EKF Jacobian
Arrhenius	Temperature T	$R_0(T), R_1(T), R_2(T)$
RC branch (ZOH)	Current i, R_ℓ, τ_ℓ	$V_{\mathrm{RC1}}, V_{\mathrm{RC2}}$
Terminal voltage	$V_{\mathrm{oc}}, V_{\mathrm{RC1}}, V_{\mathrm{RC2}}, i, R_0$	Measurable output; EKF innovation
Thermal model	Current i, R_{eff}, T	Temperature T
Arrhenius	Updated T	Updated resistances (loop back)
Capacity fade	Cumulative Ah throughput, T	Updated Q_{nom}
EKF	All of the above + measured V_t	Corrected $\hat{z}$

The complete picture. At every time step: (1) measure current and voltage; (2) update SoC via Coulomb counting; (3) evaluate Arrhenius-corrected resistances at the current temperature; (4) update RC states via ZOH; (5) compute predicted terminal voltage; (6) update cell temperature via the thermal model; (7) run the EKF to correct SoC using the voltage residual; (8) periodically update Q_{nom} using the fade model. This loop runs every 0.1–10 seconds in a real BMS.